

Tharun Derangula

Full-Stack Software Engineer | AI/ML Integration | Distributed Systems

+1 (408) 652-9472 | dtharun209@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#) | San Francisco Bay Area, California

SUMMARY

Full-stack engineer with 4+ years building and scaling production systems across frontend, backend, and cloud. Strong in distributed systems, event-driven services, and high-traffic APIs serving millions of users, with recent focus on AI/ML features using RAG, vector retrieval, and LLM integration. Owns features end to end, mentors engineers, and ships in fast Agile teams.

TECHNICAL SKILLS

Languages: Java, TypeScript, JavaScript (ES6+), Python, Go, SQL

Frontend: React, Next.js (SSR), Redux, Tailwind CSS, HTML5, CSS3, Jest, React Testing Library, Cypress

Backend: Node.js, Express.js, Spring Boot, REST APIs, GraphQL, gRPC, Microservices, Kafka, Redis

AI/ML: LLM Integration, RAG, pgvector, Vector Retrieval, LangChain, OpenAI API, Anthropic API, Prompt Engineering

Cloud & DevOps: GCP, AWS (EC2, S3, CloudWatch, DynamoDB), Docker, Kubernetes, Terraform, Helm, GitHub Actions, Prometheus, Grafana

Databases: PostgreSQL, MongoDB, Redis, Snowflake, Elasticsearch, DynamoDB

Practices: System Design, Distributed Systems, Event-Driven Architecture, Domain-Driven Design, Microservices, TDD, CI/CD, Agile/Scrum

AI Dev Tools: Claude Code, GitHub Copilot, Cursor, Codex

EXPERIENCE

Software Engineer, Walmart Global Tech — USA

Apr 2025 – Present

- Shipped customer-facing features for **Sparky**, Walmart's AI shopping agent, delivering conversational shopping and cart workflows across web and mobile at Walmart scale by building with **React, Next.js, and TypeScript** alongside PM, design, and ML teams.
- Cut p95 API latency by **35%** on high-traffic recommendation endpoints by designing **Node.js, Spring Boot, GraphQL, and REST** services with **Redis** caching and connection pooling.
- Processed millions of shopping events reliably by architecting cart auto-build and intent-routing **microservices** in **Java, Spring Boot, Kafka, and PostgreSQL** with idempotent retries and dead-letter queues, plus near real-time **GCP Pub/Sub** pipelines for inventory and fulfillment.
- Improved AI recommendation relevance by **20%** in A/B testing by integrating Walmart's **Wallaby LLM** via the Element ML Platform using **RAG, pgvector, and vector retrieval**.
- Held sub-50ms median latency through seasonal and Rollback traffic spikes by tuning **PostgreSQL** indexes and query plans and layering **Redis** caching across read-heavy services.
- Cut deploy effort by **50%** and enabled zero-downtime releases by automating infrastructure with **Docker, Kubernetes, Terraform, WCNP, and GitHub Actions**, with **Datadog and Grafana** observability.
- Held **85%+** test coverage and shortened delivery cycles by writing **Jest and React Testing Library** suites and adopting **Claude Code and GitHub Copilot** for scaffolding, refactors, and test generation.
- Sustained high-traffic shopping events without degradation by designing stateless, horizontally scalable services with connection pooling, **Redis-backed** caching, and graceful degradation patterns.
- Secured customer-facing APIs by implementing **JWT** and **OAuth 2.0** authentication with role-based access control across shopping and cart endpoints.

Software Engineer, Kennesaw State University — USA

Jan 2024 – Dec 2024

- Consolidated Owl Express, D2L Brightspace, DegreeWorks, and KSUMail into one responsive platform serving students and faculty by building KSU's **Campus Hub** with **React, Next.js, and TypeScript**.
- Improved dashboard load time by **30%** by developing **Node.js, Spring Boot, REST, and Redis** APIs and integrating **Ellucian Banner** and DegreeWorks with query tuning and caching.
- Cut manual data correction by **40%** across connected systems by owning **Java and Spring Boot** services over **PostgreSQL** with REST-webhook reconciliation for enrollment, grades, and financial aid data.
- Reduced deployment cycle time by **50%** by deploying on **AWS EC2, S3, Docker, and GitHub Actions** CI/CD with low-disruption releases during peak registration periods.
- Caught regressions earlier by maintaining **80%+** coverage with **Jest and React Testing Library** and collaborating with UITS teams in **Agile** sprints through design and code reviews.

Software Engineer, HCL Tech — India

May 2020 – Dec 2022

- Enabled invoice tracking and payment workflows for enterprise banking customers across regions by building responsive **React, TypeScript, and JavaScript** interfaces backed by **Java and Spring Boot** APIs.
- Improved peak-period API responsiveness by **35%** by optimizing **Java and Spring Boot** services on **PostgreSQL** and **Redis** with caching and connection pooling.
- Ensured reliable, exactly once transaction handling under high load by engineering event-driven payment processing with **Kafka, Java, and Spring Boot** using idempotent retries and dead-letter queues.
- Cut batch processing time by **45%** for analytics and compliance reporting by building **ETL** pipelines with **Python and GCP BigQuery** for downstream finance teams.
- Took releases from days to hours and cut production defects by **40%** by modernizing legacy backends into **microservices** on **Docker, Terraform, GCP Cloud Run, and GKE** with **GitHub Actions** CI/CD and **TDD** (JUnit, pytest).
- Improved delivery consistency across the payments platform by driving code reviews, **API design** discussions, and **Agile** delivery planning with cross-functional engineering teams.

EDUCATION

Master of science, Information Technology | Kennesaw State University

2023 – 2024

Bachelor of science, Computer Science & Engineering | Lovely Professional University

2018 – 2022

CERTIFICATIONS

- Google Professional Cloud Architect
- Google AI Essentials Specialization
- AWS Certified Solutions Architect